Weather-Conditioned Deep Learning for Bird Migration Trajectory Forecasting

Xinyan Cai, Jiayi Chen
Department of Electrical and Computer Engineering
University of California, San Diego
{xic116, jic101}@ucsd.edu
Team 53

Abstract

This report studies next-day bird trajectory prediction from historical movement records and optional weather covariates. We emphasize two methodological questions: whether a triline decomposed prediction structure improves over direct coordinate regression, and whether a Transformer encoder improves over simpler sequence and feed-forward baselines. Each model is evaluated at input context lengths $k \in \{7, 14, 30\}$ using GPS prediction error. We also compare feature settings with and without weather data to measure whether aligned environmental covariates improve forecasting accuracy.

1 Introduction

Bird migration forecasting is a spatiotemporal sequence prediction problem. Given the previous k daily observations for an individual bird, the goal is to predict the next-day GPS location. This task is challenging because many days contain little movement, while migration days can involve large displacements whose direction and distance depend on recent trajectory history, seasonal timing, and environmental conditions.

This project focuses on three comparisons:

1. **Direct vs. Triline:** direct regression predicts the next displacement vector, while the triline model predicts fly/no-fly, movement distance, and heading separately before reconstructing the next location.
2. **Transformer vs. baselines:** a 2-layer Transformer is compared with a 2-layer LSTM and a flattened-window MLP baseline.
3. **Weather vs. no weather:** models trained with weather covariates are compared against models trained only on trajectory and temporal features.

2 Data

2.1 Dataset

TODO: Describe the final dataset used in the report: source, filtering rules, date range, number of birds, number of paths, and number of daily observations.

Preprint.

Table 1: Dataset summary.

Field	Value
Daily records	TODO: fill
Birds	TODO: fill
Migration paths	TODO: fill
Date range	TODO: fill
Train/test split	TODO: fill
Prediction target	Next-day latitude and longitude
Context lengths	$k = 7, 14, 30$

2.2 Features

The no-weather setting uses trajectory, movement, identity, and time features such as latitude, longitude, displacement, step length, heading, day-of-year encoding, and stopover duration. The weather setting augments these features with aligned environmental covariates such as wind, pressure, precipitation, temperature, or derived tailwind/crosswind variables.

Table 2: Feature groups used by each experiment setting.

Feature group	No weather	With weather
Location and movement	Yes	Yes
Temporal encoding	Yes	Yes
Bird identity embedding	Yes	Yes
Weather covariates	No	Yes

3 Methods

3.1 Direct Regression Model

The direct model treats trajectory prediction as coordinate regression. For each input window, the model encodes the past k daily feature vectors and predicts the next displacement:

$$\widehat{\Delta \mathbf{x}}_{t+1} = [\widehat{\Delta \mathrm{lat}}_{t+1}, \widehat{\Delta \mathrm{lon}}_{t+1}].$$

The predicted GPS location is reconstructed as

$$\widehat{\mathbf{x}}_{t+1} = \mathbf{x}_t + \widehat{\Delta \mathbf{x}}_{t+1}.$$

This structure is simple and directly optimized for coordinate error, but it does not expose intermediate movement decisions such as whether the bird flies.

3.2 Triline Decomposed Model

The triline model decomposes movement into three interpretable prediction heads:

$$\widehat{p}_{fly} = f_{fly}(\mathbf{h}), \tag{1}$$

$$\widehat{d} = \exp(f_{dist}(\mathbf{h})) - 1, \tag{2}$$

$$[\widehat{\sin\theta}, \widehat{\cos\theta}] = f_{dir}(\mathbf{h}). \tag{3}$$

The predicted distance and heading are converted back into latitude and longitude displacement. This design separates the migration decision, movement magnitude, and movement direction. It can also support optional fly/no-fly gating, where predicted distance is set to zero when $\widehat{p}_{fly}$ is below a threshold.

3.3 Transformer Encoder

The main model uses a 2-layer Transformer encoder. Each daily feature vector is projected into a hidden dimension, combined with positional information and bird identity embedding, then processed with self-attention:

$$\mathbf{H} = \mathrm{TransformerEncoder}(\mathbf{z}_{t-k+1}, \ldots, \mathbf{z}_t).$$

The final representation is used either by the direct coordinate head or by the triline heads. The Transformer is expected to help when long-range interactions across the input window matter.

3.4 Baseline Models

We compare the Transformer against two learned baselines:

- **2-layer LSTM:** processes the input sequence recurrently and predicts from the final hidden state.
- **MLP:** flattens the k-day input window and predicts with feed-forward layers.

Classical baselines such as persistence and constant velocity may also be reported as reference points.

4 Experimental Setup

Table 3: Common training configuration.

Parameter	Value
Context lengths	$k = 7, 14, 30$
Transformer depth	2 layers
LSTM depth	2 layers
Optimizer	AdamW
Learning rate	TODO: fill
Batch size	TODO: fill
Early stopping	TODO: fill
Random seed	TODO: fill

4.1 Metrics

The primary metric is haversine GPS error in kilometers between predicted and true next-day locations. We report mean error for overall accuracy, median error for typical-day behavior, and optionally P90 or migration-day error to measure difficult large-movement cases.

5 Results

5.1 Comparison 1: Direct vs. Triline

This comparison isolates the prediction structure. Both methods should use the same encoder family and the same feature setting when possible. The headline question is whether decomposing movement into fly/no-fly, distance, and heading improves GPS error relative to direct coordinate regression.

Table 4: Direct regression vs. triline decomposition across context lengths. Lower GPS error is better.

Model	k	Mean km	Median km	P90 km	Migration mean km	Notes
Direct Transformer 2L	7	TODO:	TODO:	TODO:	TODO:	TODO:
Triline Transformer 2L	7	TODO:	TODO:	TODO:	TODO:	TODO:
Direct Transformer 2L	14	TODO:	TODO:	TODO:	TODO:	TODO:
Triline Transformer 2L	14	TODO:	TODO:	TODO:	TODO:	TODO:
Direct Transformer 2L	30	TODO:	TODO:	TODO:	TODO:	TODO:
Triline Transformer 2L	30	TODO:	TODO:	TODO:	TODO:	TODO:

Plot placeholder

Filename listed in caption.

Replace with generated error plot.

Figure 1: Direct vs. triline GPS error for $k = 7, 14, 30$. Recommended plot: grouped bar chart or line plot with one line per method and mean GPS error on the y-axis.

5.2 Comparison 2: Transformer vs. LSTM and MLP Baselines

This comparison isolates the sequence model architecture. Use the triline prediction structure for Transformer and LSTM if the goal is to compare encoders under the same output formulation. Use a flattened-window MLP as the non-sequential learned baseline.

Table 5: 2-layer Transformer vs. 2-layer LSTM and MLP baselines across context lengths. Lower GPS error is better.

Model	k	Mean km	Median km	P90 km	Parameters
Triline Transformer 2L	7	TODO:	TODO:	TODO:	TODO:
Triline LSTM 2L	7	TODO:	TODO:	TODO:	TODO:
MLP	7	TODO:	TODO:	TODO:	TODO:
Triline Transformer 2L	14	TODO:	TODO:	TODO:	TODO:
Triline LSTM 2L	14	TODO:	TODO:	TODO:	TODO:
MLP	14	TODO:	TODO:	TODO:	TODO:
Triline Transformer 2L	30	TODO:	TODO:	TODO:	TODO:
Triline LSTM 2L	30	TODO:	TODO:	TODO:	TODO:
MLP	30	TODO:	TODO:	TODO:	TODO:

Plot placeholder

Filename listed in caption.

Replace with generated error plot.

Figure 2: Transformer vs. LSTM and MLP error across $k = 7, 14, 30$. Recommended plot: grouped bar chart with one group per k and one bar per architecture.

5.3 Comparison 3: With Weather vs. Without Weather

This comparison measures the value of environmental covariates. Use the same model class, output structure, context length, split protocol, and evaluation metric in both settings. If possible, report both mean GPS error and migration-day error, since weather may be more informative on large flight days than on stationary days.

Table 6: Weather ablation across context lengths. Lower GPS error is better.

Model	k	No-weather mean	Weather mean	Delta	No-weather median	Weather median
Triline Transformer 2L	7	TODO:	TODO:	TODO:	TODO:	TODO:
Triline Transformer 2L	14	TODO:	TODO:	TODO:	TODO:	TODO:
Triline Transformer 2L	30	TODO:	TODO:	TODO:	TODO:	TODO:
Triline LSTM 2L	7	TODO:	TODO:	TODO:	TODO:	TODO:
Triline LSTM 2L	14	TODO:	TODO:	TODO:	TODO:	TODO:
Triline LSTM 2L	30	TODO:	TODO:	TODO:	TODO:	TODO:

Plot placeholder

Filename listed in caption.

Replace with generated error plot.

Figure 3: With-weather vs. no-weather GPS error across $k = 7, 14, 30$. Recommended plot: paired bars for each model and context length, or separate line plots for no-weather and weather settings.

6 Discussion

6.1 Direct vs. Triline

TODO: Summarize whether triline improves mean, median, or migration-day error. Discuss whether the fly/no-fly head improves interpretability or enables useful gating. Also mention cases where direct regression performs better, especially on large movement days if observed.

6.2 Transformer vs. Baselines

TODO: Discuss whether self-attention improves over the 2-layer LSTM and MLP. If gains are small, explain that recent movement features or dataset size may dominate architecture choice.

6.3 Effect of Weather

TODO: Discuss whether weather features improve error. If weather does not improve results, consider possible reasons: noisy joins, weather variables not aligned with departure decisions, insufficient capacity, or train/test distribution shift.

7 Conclusion

TODO: State the final empirical takeaways in 3–5 sentences. Explicitly answer: (1) direct or triline, (2) Transformer or baseline, and (3) weather or no weather.

Reproducibility Checklist

Table 7: Artifacts to report for reproducibility.

Item	Location or value
Dataset path	TODO: fill
Training script	TODO: fill
Result CSV files	TODO: fill
Figure folder	`figures/`
Random seed	TODO: fill
Hardware	TODO: fill